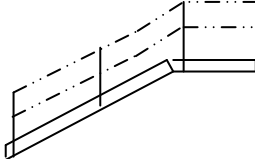
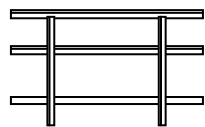


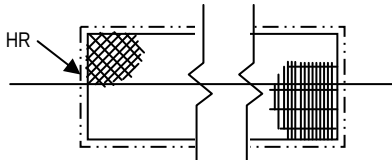


STRUCTURAL STEEL – ESTIMATING WEIGHTS

	Stair weights shown Include: A. C10x15.3 Stringer B. 1-1/4 x 3/16 Bearing Bar Grating Treads 10-7/8" Wind w/Chkd. PL Nosing C. Standard Handrail	Stair weight – Lbs/Ft of rise					
			Code	Width	Rise/Tread	HR Type	Weight
		Std As Req'd.	OSHA/BOCA	2'-6"	7-3/4"	Angle	127 #
					9-3/4"	Pipe	111 #
			Std. Bldg.	3'-0"	7-3/4"	Angle	135 #
					9-3/4"	Pipe	118 #
Life Safety	3'-0"	7"	Angle	160 #			
			11"	Pipe	141 #		

	Handrail weights shown include: Angle handrail – L 2-1/2 x 2-1/2 x 1/4 toprail, midrail, posts and toe PL 1/4 x 4 Pipe handrail – 1-1/4 Sch 40 pipe toprail, midrail and posts and toe PL 1/4 x 4 Note: Handrail weights are based on 6'-0" post spacing.	Handrail	
		HR Type	Weight
		Angle	15.0 #/ft
		Pipe	10.0 #/ft

	Ladder weights shown include: Side rails, rungs and miscellaneous ladder braces. Ladder and cage weights shown include: Same as above plus vertical cage bards and hoops at 4'-0" spacing. Note: Add 7.1 # for safety gate at each ladder.	Ladder Only lbs/ft		Ladder and Cage lbs/ft	
		Side Rail	Weight	Side Rail	Weight
		Bar 2-1/2 x 3/8	8.5	Bar 2-1/2 x 3/8	20.0
		Bar 3 x 3/8	10.0	Bar 3 x 3/8	21.5
		C3 x 4.1	10.5	C3 x 4.1	22.0

	Weights shown include: framing, covering and handrail.	Platform Area	Platf. w/ 1/4" chk'd PL covering Lbs/Sq Ft	Platf. w/ 1-1/4 x 3/16 grating covering Lbs/Sq Ft
		< 50 Sq. Ft.	30	28
		50 to 100 Sq. Ft.	25	23
		> 100 Sq. Ft.	22	19

	Checkered Floor Plate	1/4" = 11.25 lbs / sq. ft 5/16" = 13.80 lbs / sq. ft. 3/8" = 16.37 lbs / sq. ft. 1/2" = 21.47 lbs / sq. ft.			
	1-1/4" x 3/16" Bearing Bars at 1-3/8" c/c = 9.2 lbs / sq. ft.				
	Structural Steel (Domestic)	Add for connections	Non Seismic	Seismic	
		Light	Under 20 lbs	15 %	15 %
		Medium	Over 20 to 40 lbs	10 %	13 %
		Heavy	Over 40 to 60 lbs	5 %	7 %
		Ex Heavy	Over 60 lbs	3 %	4 %